

ORIGINAL ARTICLE

Best Practices for Leveraging PISA CT Data to Understand Gender Differences in Creative Thinking

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ABSTRACT

Access to data from the 2022 PISA creative thinking assessment (PISA CT) provides a unique opportunity to advance our understanding of gender differences in creativity. However, in addition to the general theoretical and methodological considerations discussed elsewhere in this special issue, there are several matters specific to gender differences in creativity that should be considered when interpreting the results of the PISA CT. First, the overall creative thinking index on the PISA CT may not provide much value to understanding gender differences in creativity, as differences may be domain and task specific. Second, gender differences in careless responding may bias results for gender differences in PISA CT scores, as effort is associated with enhanced creative performance. Third, the restricted age range of PISA participants may limit the generalizability of results, as the developmental dynamics of gender differences in creativity suggest that the size and direction of gender differences in creative performance may change during this stage of adolescence. Each of these issues, as well as potential solutions and directions for future research using PISA CT scores to examine gender differences in creativity, are discussed.

Researchers have long maintained that differences in the creative ability and potential of men/boys and women/girls at the little-c level (everyday, nonexpert creative activities; Kaufman and Beghetto 2009) tend to be minimal overall (see Baer and Kaufman 2008; Kogan 1974). This assertion has been supported by recent meta-analyses of studies in which performance on a creative task is measured (Alabbasi et al. 2025; Taylor et al. 2024). However, there are still limitations to our understanding of gender differences in creativity at this level (Taylor 2023). For example, the majority of studies examining gender differences in creativity have come out of Anglo countries, particularly the United States,¹ and have used divergent thinking tasks as indicators of creativity (see Alabbasi et al. 2025; Taylor et al. 2024). Access to data from the 2022 PISA creative thinking assessment (PISA CT)—with tasks representing different domains and creative skills from participants across a wide variety of countries—provides a unique opportunity to advance our understanding of gender differences in creativity.

The PISA CT report of findings (OECD 2024b) dedicates a fair amount of space to describing differences between boys and girls across the participating countries and economies. Conclusions based on the findings align with existing research on gender differences in creativity in some ways, yet not others. On the PISA CT, girls outperformed boys across all creative thinking tasks in average performance. The average score-point gender difference reported by the PISA CT of 3 points translates to a difference between boys and girls of 0.30 of a standard deviation, yet some countries saw over a 0.50 SD advantage in girls' average scores (OECD 2024b). Although meta-analyses suggest that women and girls have a slight advantage in performance on tests of creative ability on average, effect sizes have been smaller than that found on the PISA CT (≤ 0.10 SD difference; Alabbasi et al. 2025; Taylor et al. 2024).

On the PISA CT, boys were also less likely than girls to be in the 75th percentile of scorers within their country or region. This

finding conflicts with the Greater Male Variability Hypothesis in creativity, which suggests that men/boys vary more than women/girls in creative ability, resulting in more men/boys scoring in the upper (and lower) extremes of creative ability (He and Wong 2011).² Though meta-analysis shows gender differences in variability to be trivial (Taylor et al. 2024), boys tend to be overrepresented at the upper extremes of a distribution of creativity scores on at least some tasks across individual studies (Karwowski et al. 2016; Taylor and Barbot 2021b). Understanding the meaning of these discrepancies requires a discussion of the PISA CT findings in the context of existing research in the field.

1 | Considerations for Gender Differences in PISA CT Scores

There are quite a few general theoretical and methodological issues to consider when using data from the PISA CT (see Barbot and Kaufman 2025). However, there are also several matters specific to gender differences in creativity that should be considered. These include gender differences in effort and careless responding, domain and task specificity, and the developmental dynamics of gender differences in creativity.

1.1 | Gender Differences in Effort and Careless Responding

Gender differences in engagement were found across PISA CT tasks, wherein boys had more rapid responses (i.e., less than 30s with no response or a no-credit response), relative rapid responses (i.e., lower 25% of response time with no response or a no-credit response), or provided no response more than girls (OECD 2024b). This is consistent with self-reported effort on the 2022 PISA overall (OECD 2024a), as well as previous work showing that girls and women tend to exert more effort on tests than boys and men, particularly in low-stakes testing conditions (DeMars et al. 2013). Research has shown that girls also sustain test performance more than boys over longer testing periods, such as the 2-h-long PISA testing session; for example, girls in most countries exhibited less performance decline (i.e., the increasing likelihood of answering questions incorrectly as one moves toward the end of a test) than boys across reading, math, and science domains on the 2009 PISA (Balart and Oosterveen 2019). These types of gender differences in effort may bias results for gender differences in PISA CT scores, as beyond the necessity of providing a response for credit, effort is associated with enhanced creative performance. Generating creative responses in a testing situation requires sustained focus, persistence, and engaging deeply enough to avoid providing the first, superficial response (Benedek et al. 2012; Silvia and Beaty 2012; Taylor et al. 2020).

Gender differences in effort may be particularly problematic when paired with the PISA CT scoring scheme, in which “In general, students who invested a minimum level of effort in a task and who suggested an idea implicitly or explicitly connected to the task stimulus were awarded partial credit” (OECD 2024b, 133). This means that, because those who made any effort to respond to the task appropriately received partial credit, girls

would have been more likely to receive credit even for noncreative responses.³ Though girls also received a greater proportion of full-credit responses than boys, discounting any partial credits, it is unclear how many responses from boys did not receive credit due to insufficient creativity versus because a response was not attempted.

Indeed, variation in gender differences in creative thinking across the countries and economies appears to correspond quite clearly with indicators of engagement on the PISA CT assessment. For example, the only countries in which gender differences were not significant for the overall creative thinking index (Chile, Mexico, and Peru) were also three of the countries that had the lowest percentage point differences in disengagement factors. Additionally, the gender gap in performance was strongest in the written expression domain and weaker in the scientific problem-solving domain, mirroring the gender gap in indicators of engagement (OECD 2024b). This suggests that the results may reflect differences in effort, as opposed to creative potential or readiness outside of the testing context.

1.2 | Domain and Task Specificity

The overall creative thinking index on the PISA CT may not provide much value to understanding gender differences in creativity, as differences may be domain and task specific. Theory and research suggest that creative performance is comprised of both a general ability and domain and task-specific knowledge and skills (Baer and Kaufman 2005; Barbot et al. 2016). Indeed, though girls outperformed boys in each of the four domains included on the PISA CT (written expression, visual expression, social problem-solving, and scientific problem-solving) on average, there was a fair amount of variation in domain patterns across the countries. Girls significantly outperformed boys in most countries in written and visual expression, but there were no significant gender differences in most countries in scientific problem-solving. There were only nine countries in which girls significantly outperformed boys in all four domains (Bulgaria, Denmark, Finland, Iceland, Jordan, Malaysia, Qatar, Saudi Arabia, and United Arab Emirates). The remaining 54 countries demonstrated different patterns of gender differences across domains. For example, in Mexico, girls significantly outperformed boys in visual expression, boys significantly outperformed girls in scientific problem-solving, and there were no gender differences in written expression or social problem-solving. However, in Costa Rica, girls significantly outperformed boys in visual expression and social problem-solving, and there were no gender differences in written expression or scientific problem-solving (OECD 2024b).

Larger or more consistent gender differences may be found in domains not included on the PISA CT. It is understandable that, given the time-limited nature of the test and considerable resources required for scoring a large number of responses, there were a limited number of domains included on the PISA CT. The inclusion of written expression and scientific problem-solving creative tasks in particular aligns with recent research demonstrating domain differences in gendered attributions of creativity, wherein people believe that women are more creative in literature and men are more creative in science (Taylor and

Barbot 2023). However, because a consistent pattern of gender differences in performance in creative domains and tasks has not been identified by research (Abraham 2016; Taylor et al. 2024), it is difficult to determine whether or not a different set of domains or tasks would result in a different pattern of gender differences.

The method of data collection might have also obscured some of the reasons domain differences may emerge. Drawing on the specialization-differentiation hypothesis (Barbot and Tinio 2015), variability in gender differences across domains may emerge due to gender-specific specialization that impacts skills and abilities not directly related to creativity itself but that are necessary for creativity in a given domain. For example, though not well supported by research, boys are frequently thought to perform better on visual-figural creativity tasks than girls (Abraham 2016). Two of the reasons that boys have been suggested to outperform girls on visual-figural tasks are that (1) girls have inferior visual-spatial abilities compared to boys (Abra and Valentine-French 1991) and (2) girls make less socially valued stylistic choices (e.g., colors and shapes) in their creative products than boys (Harris 1989). Assuming that these assertions were true and would lead to boys outperforming girls on visual-figural creativity tasks, these differences would likely be obscured when making collages in which the background and elements are provided, as opposed to more real-world creations. Thus, the PISA CT may have missed some gender-based variations by limiting the influence of domain-specific skills and real-world creative processes.

Creative tasks across all domains required reading comprehension, an area in which girls outperformed boys (OECD 2024a), and the mode of expression for three out of the four domains was written. This meant that regardless of the domain, students had to articulate their thoughts in writing, which may have amplified the influence of reading and writing skills on scores. The report of results highlights this dynamic, suggesting that “Even after accounting for reading performance – and despite similar gender differences observed in reading performance in general – girls’ performance advantage in creative thinking remains statistically significant in around half of all countries/economies” (OECD, 2024, 111). In other words, gender differences in overall creative performance are *no longer statistically significant* after accounting for reading performance in around half of all countries/economies. This suggests that tasks on the PISA CT may have inadvertently favored reading and writing skills, skewing results in girls’ favor regardless of the domain.

1.3 | Developmental Dynamics of Gender Differences in Creativity

Though the restricted age range of PISA participants (i.e., 15 years and 3 months to 16 years and 2 months old) allows for meaningful cross-sectional comparisons across the participating countries, it also limits the generalizability of the findings. This may be particularly true for gender differences in creative performance. Research suggests that similar to other cognitive abilities (e.g., Arden and Plomin 2006), the magnitude and direction of gender differences in creative performance may change during this stage of adolescence (He 2018; He et al. 2015; Said-Metwaly et al. 2021). For example, He (2018) conducted

a longitudinal study in which participants in three age groups (ranging from 8 to 18 years old at Time 1) completed the Test for Creative Thinking-Drawing Production (Urban and Jellen 1996) annually over 4 years. The findings revealed that female participants outperformed male participants in mean creativity for age groups under 16. However, for participants aged 16 and older, there were no significant gender differences in mean creativity. Furthermore, though male participants were significantly overrepresented in the lower tails of the distribution of scores in all age groups, they were overrepresented in the upper tails of the distribution only in age groups over 16. Therefore, the utility of the PISA CT results for understanding gender differences in creativity in other age groups is uncertain.

2 | Guidelines for Best Practices

To summarize, there are at least three issues with the PISA CT that should be accounted for when conducting secondary analyses using these data to understand gender differences in creativity. First, gender differences in engagement on the PISA CT tasks suggest that girls’ greater effort may have biased results, reflecting gender differences in effort rather than creative potential. Second, aggregating data for domains and tasks included on the PISA CT may have obscured gender-based variations in domain patterns across countries and economies. Third, the restricted age range of PISA participants limits the generalizability of findings for gender differences in creativity, as research shows that these differences may shift in later adolescence. Based on these issues, there are several suggestions for best leveraging PISA CT data to understand gender differences in creative thinking.

2.1 | Control for Gender Differences in Effort

Perhaps, the most important step to take when working with PISA CT data to examine gender differences is to control for gender differences in effort. There are at least three solutions for correcting for effort. First, the simplest solution would be to exclude all data for participants who exhibit low effort. However, although this solution is feasible given the large sample size of participants, listwise exclusion may induce selection bias, as low effort may in fact be related to creative ability (Rios et al. 2017). Further, excluding participants exhibiting low effort would likely result in an underrepresentation of boys. Second, individual items identified as rapid responses (and nonresponses) can be treated as missing data (i.e., effort-moderated scoring; Wise and Demars 2006). This method requires the missing responses to be missing at random (MAR), which allows for missingness to be related to the outcome if the inclusion of other variables in the model eliminates the relationship (DeMars 2024). Third, effort can be included in measurement models as a control variable. This method allows researchers to include additional indicators of effort (e.g., length of written responses or time-on-task; Taylor and Barbot 2021a), demographic variables, and/or scores on other PISA components (e.g., reading, math, or science, all of which were correlated with CT scores at ≥ 0.66) as control variables simultaneously.

Anaya and Zamarro (2024) used the modeling approach to examine the gender gap in math, science, and reading scores on

the 2015 PISA. After adjusting students' scores based on effort (i.e., nonresponse rates on the background survey and proportion of rapid guessing across the entire assessment), they found that the estimated gender gap was wider in the case of math and science and narrower in the case of reading than when unadjusted scores were used. For example, in some countries, boys outperformed girls on the science module up to 40% of a standard deviation greater after adjusting for effort. Given that girls outperformed boys on the PISA CT on average and exhibited greater effort/engagement with the test, scores adjusted for effort may result in a narrowing of the observed gender gap in creative performance. In any case, accounting for gender differences in effort is likely to provide a more accurate portrayal of any differences in girls' and boys' creative performance.

It may also be informative to examine *when* boys exhibit greater disengagement than girls on the creative tasks. Because student assignment to different test forms of the PISA CT was randomized, effects related to the order of the presentation of items would not be apparent when aggregating across tasks or by directly comparing individual tasks. Balart and Oosterveen (2019) found that, although boys initially outperformed girls in math and science in 58 out of 74 participating countries on the 2009 PISA, girls were better able to sustain performance across all countries (i.e., performance decline was weaker for girls than boys). As a result, the ending gender gap in these modules was offset, and in some cases reversed, in more than 20% of the countries. Given that girls tend to sustain test performance longer, gender differences in effort may be smaller for tasks that appeared earlier in the testing period.

2.2 | Focus on Domain and Task-Specific Differences

Researchers should limit use of the overall creative thinking index on the PISA CT to describe gender differences in creative performance. Less than 15% (9 out of 63) of the participating countries and economies had consistent gender-based outcomes across all four domains of the PISA CT (OECD 2024b). Aggregating all four domain scores creates an inaccurate portrayal of boys' and girls' performance across the different creative tasks. Rather, given the varying domain patterns of gender differences across the participating countries and economies, it may be informative to determine what factors predict domain differences across the countries. One way to accomplish this would be to combine PISA CT data with external indicators of cultural differences in factors such as gender role norms or how creativity is valued.

The issues discussed in relation to the measurement of the different domains are common to many studies attempting to examine creativity across domains. The problem is, similar to that with gender differences in effort, these shortcomings are exacerbated by the scoring scheme of the PISA CT. Receiving partial credit for any response that is somewhat appropriate for the task means that girls were more likely to receive credit, regardless of how creative their response may or may not have been, simply because they were more likely to understand the task instructions and adequately express their ideas in writing. Controlling for reading scores may help to mitigate some of the artifacts that this scoring scheme is

likely to induce. However, whether or not to treat reading or writing ability as a confounding variable or as a natural aspect of creative production may be a matter of personal preference. After all, real-world creative performance in science does require the ability to articulate one's ideas, frequently in writing. In that case, to remediate the problems with the scoring scheme, rescoring the individual products may be the only solution. There are already efforts underway to extract scores from the PISA CT that are more consistent with the best practices in the field, such as the Consensual Assessment Technique (Amabile 1982).

2.3 | Refrain From Overgeneralizing

Given the restricted age range of participants, interpretations of gender differences in creativity in general based on PISA CT scores require extra caution. The narrow range of participants' ages may explain, at least in part, why gender differences were larger on the PISA CT than that for creative ability and potential in recent meta-analytic work (Alabbasi et al. 2025; Taylor et al. 2024), as these meta-analyses included participants of all age groups. However, as participants represent an age group for which gender differences in creativity may change rapidly (e.g., He 2018), researchers should refrain from overgeneralizing findings and/or making recommendations for other age groups based on these findings.

3 | Conclusion

The PISA CT data can be invaluable in understanding gender differences in creativity, if handled properly. This article has outlined three key guidelines specifically for using the PISA CT data to examine gender differences in creativity: (1) control for gender differences in effort, (2) focus on domain and task-specific differences, and (3) refrain from overgeneralizing. Though additional considerations (e.g., potentially controlling for reading scores and exploring cultural predictors of gender difference patterns) have been briefly discussed, the three core recommendations presented here should serve as essential principles for any future analysis in this area. Guidelines for working with PISA CT data more broadly that are discussed in other articles in this special issue should also be considered (see Barbot and Kaufman 2025). Combining the recommendations discussed provides an excellent opportunity to better understand gender differences in creativity—across domains, as well as cultural differences, and what country level factors might contribute to these differences.

Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Endnotes

¹ The Anglo cluster of countries identified in the GLOBE study includes Australia, Canada, England, Ireland, New Zealand, South Africa

(White sample), and the United States of America (House et al. 2004). Of note, the United States opted not to participate in the 2022 PISA CT module. Given the large proportion of research on gender differences in creativity coming from the United States, this may limit our understanding of how the PISA CT results relate to existing research overall.

²Of note, greater variability does not necessarily translate into overrepresentation at the upper extremes of a distribution. Though the report of results suggests that "... the overall variation in performance in creative thinking among boys is larger than among girls (Table III. B1.3.2)..." (OECD, 2024, p. 107), it is unclear what evidence supports this statement. The table readers are referred to show no difference in SD on average or for the majority of countries.

³Of note, girls may not have been more likely than boys to provide a response if poor responses were penalized (see Baldiga 2014).

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